Project Seminar





Master IET International Students SS2026

Lesson 1: Introduction

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FIW: Fachbereich Elektrotechnik und Informatik

HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

Study Regulations - University of Applied Sciences Wismar (1)

- ***Seminar implementation:***
  - The project seminar will take place **in presence in Building 1 Room 221** (otherwise you will be informed in time)

- ***Registration:***
  - register for the project seminar on **Stud.IP**, where you will also find additional information and materials

https://studip.hs-wismar.de

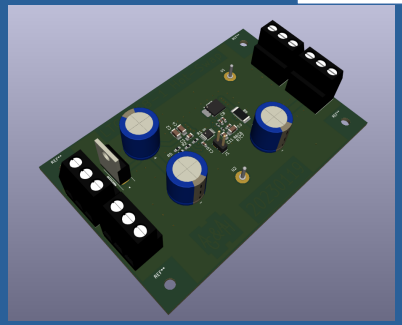
Study Regulations - University of Applied Sciences Wismar (2)

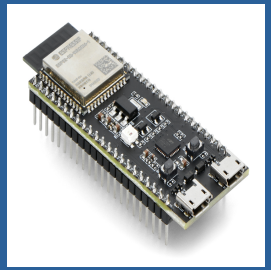
- ***Learning objectives and competences:***
  - **ability to work independently** on electrical engineering tasks with project character
  - **to work individual on projects** and practical tasks of the respective field of competence are **independently worked on**
- ***Study Exam:***
  - **Examination:** pcb project, project report (documentation) and a written test

Study Regulations (3) – Examination

PCB design

KiCad





solutions

Project documentation

Project report as PDF-file

- ***Schematics, BOM, manufacturing, assembly,***
- ***Solutions to tasks & exercises, etc.***

Put all parts together (PDF & all project releated folders) as Zip-File(s)!

Name, Matr.-No., face-photo of the student on the project cover on cover!

Study Regulations (4) - Exam

- ***Submission date of all project related files and documents:***

Saturday 20th June 2026 until 10 p.m. → deadline !

Project Seminar 2026





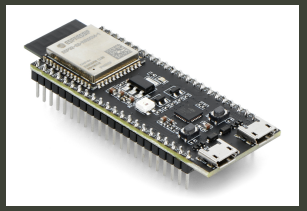
Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 1: Introduction

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Project Seminar - Content



Project Seminar

Board design in KiCad

- design pcb
- independently working on your project task

Circuit analysis

- Measurement techniques
- Operational Amplifier circuits

Software development

- become familiar with micro-controllers (µC)
- Dev. operational monitoring board software
- independently working on different programming tasks

Project Seminar - What’s needed ? (1/2)

- ***Notebook and external mouse (no touchpad !):***
  - Your notebook should use Windows 11 or Linux distributions as operating system. In case of MacOS a virtual machine from university server can be used.
  - Access to virtual machines:

IT-Info - Hochschule Wismar

Project Seminar - What's needed ? (2/2)

- ***KiCad EDA version 9.0.7 (or older stable version):***
  - You can download KiCad from:

https://www.kicad.org/

Project Seminar - What is KiCad ? (1/8)

- Open source software suite for Electronic Design Automation (EDA)
- KiCad was first released in 1992 by its original author, Jean-Pierre Charras
- It has been in continual development since then and now managed by the KiCad Development Team & it is constantly being developed
- KiCad can handle Schematic Capture and PCB Layout with Gerber output.



Project Seminar – What is KiCad ? (2/8)



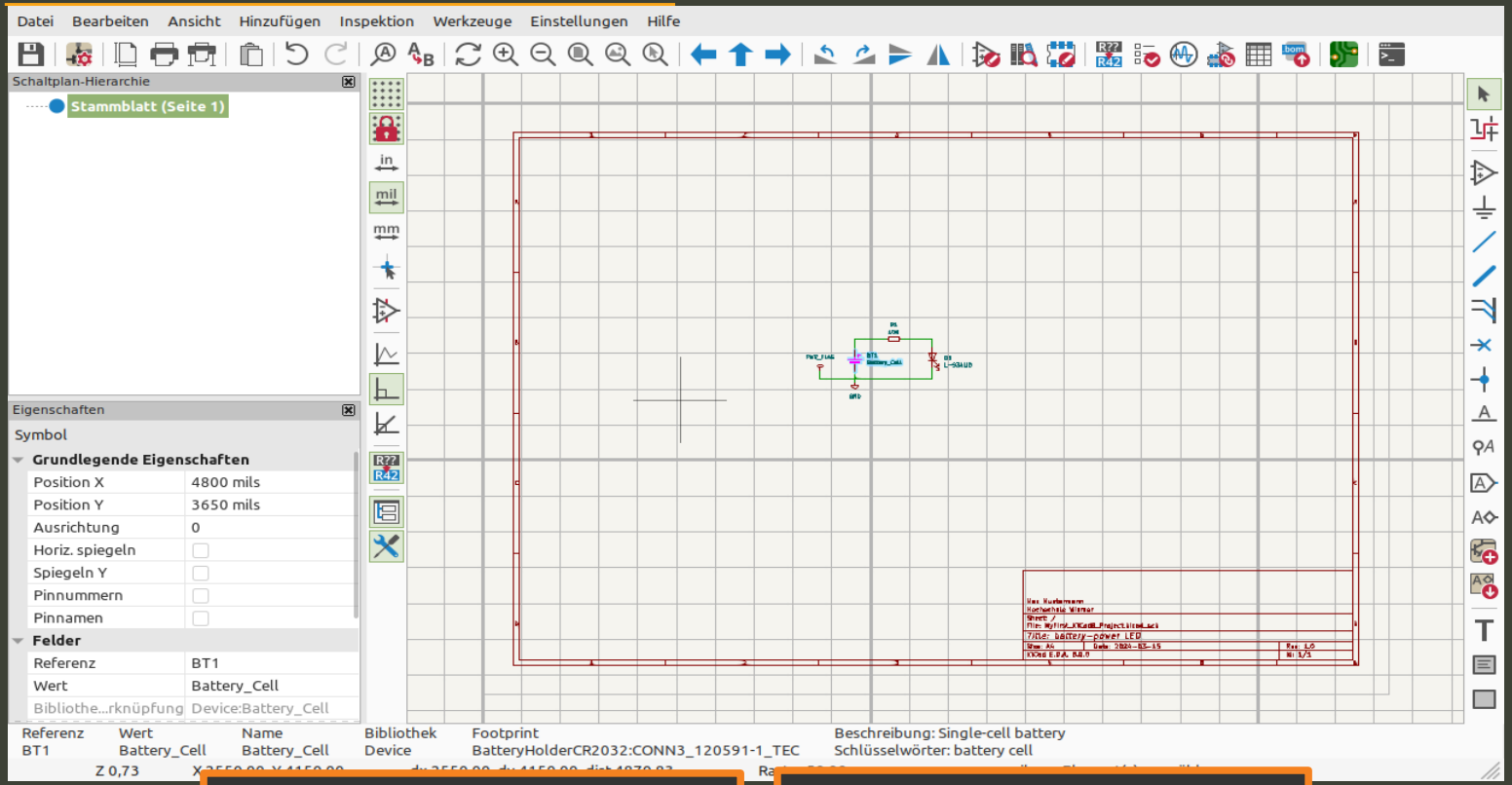
Project manager KiCad 9

https://www.kicad.org

https://docs.kicad.org/

Project Seminar – What is KiCad ? (3/8)

Schematic Editor KiCad 9

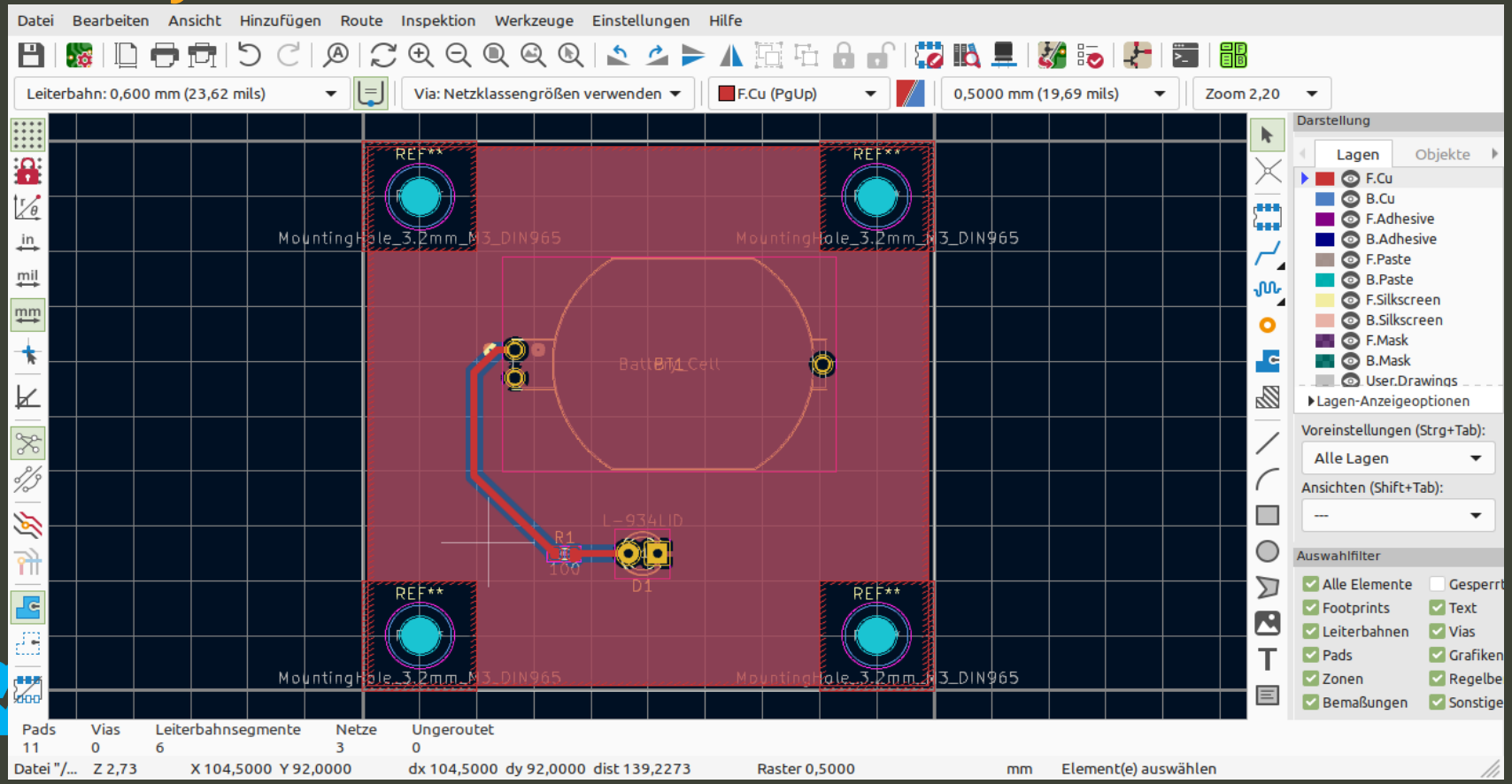


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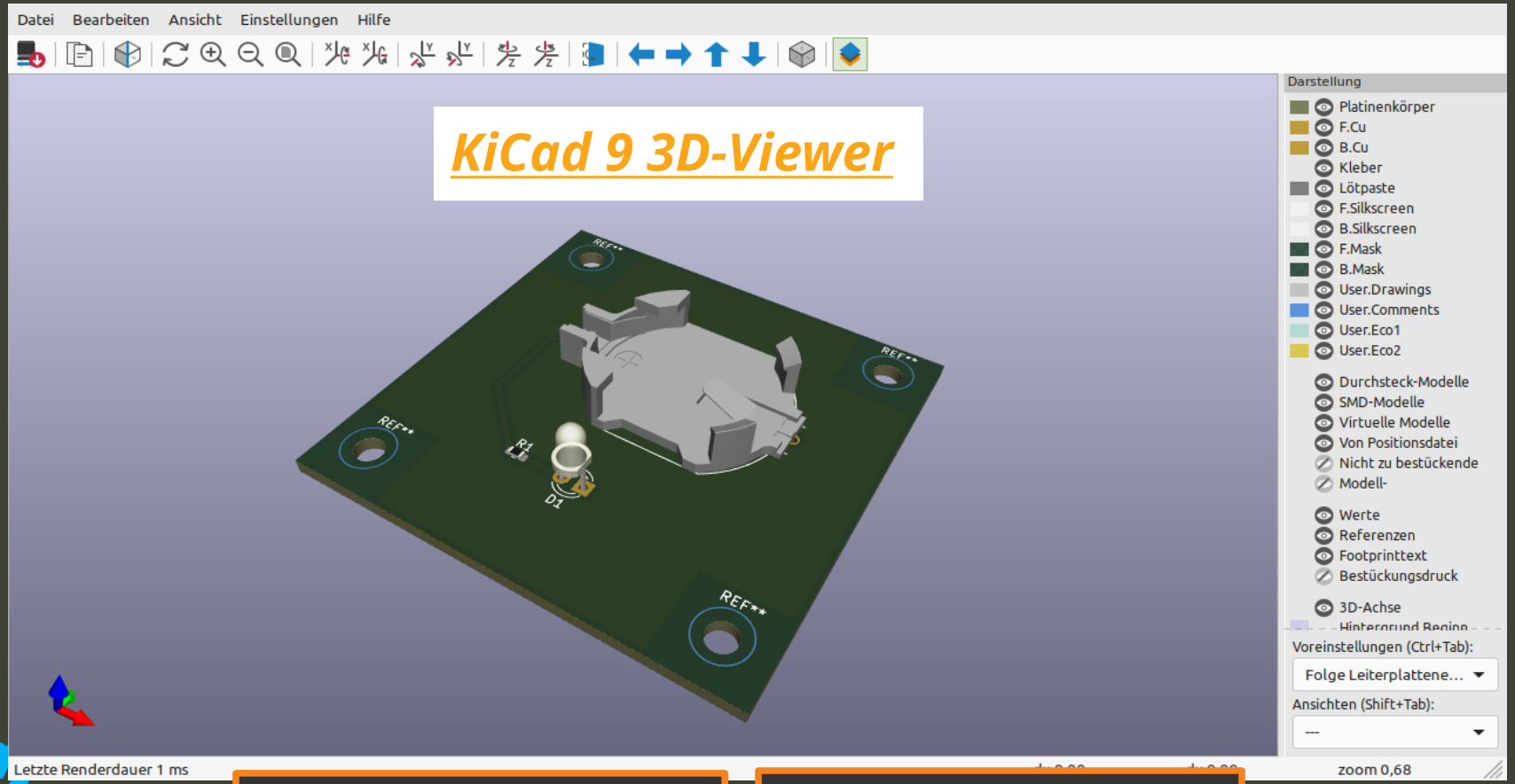
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Project Seminar – What is KiCad ? (4/8)

PCB Layout Editor KiCad 9



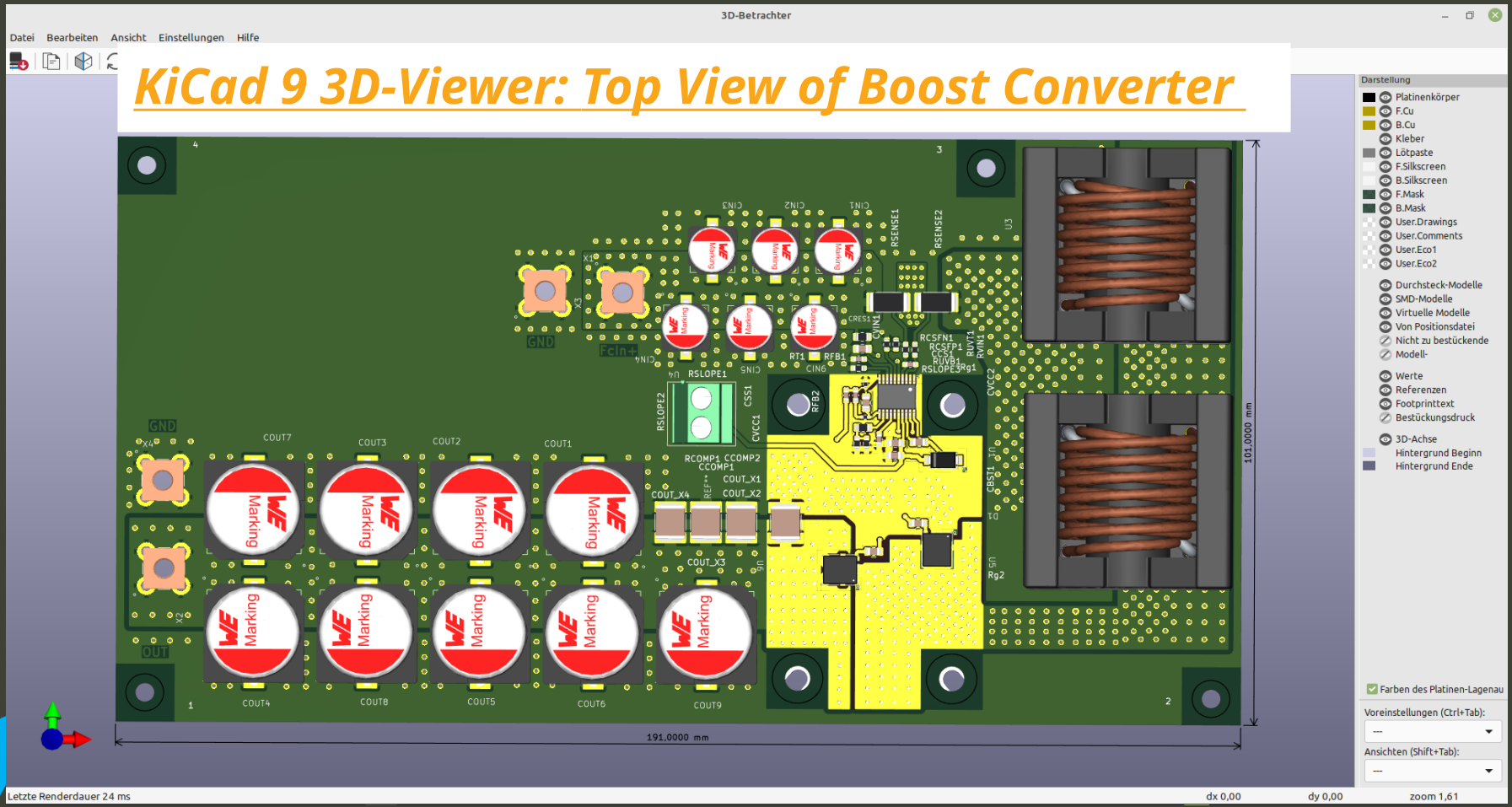
Project Seminar – What is KiCad ? (5/8)



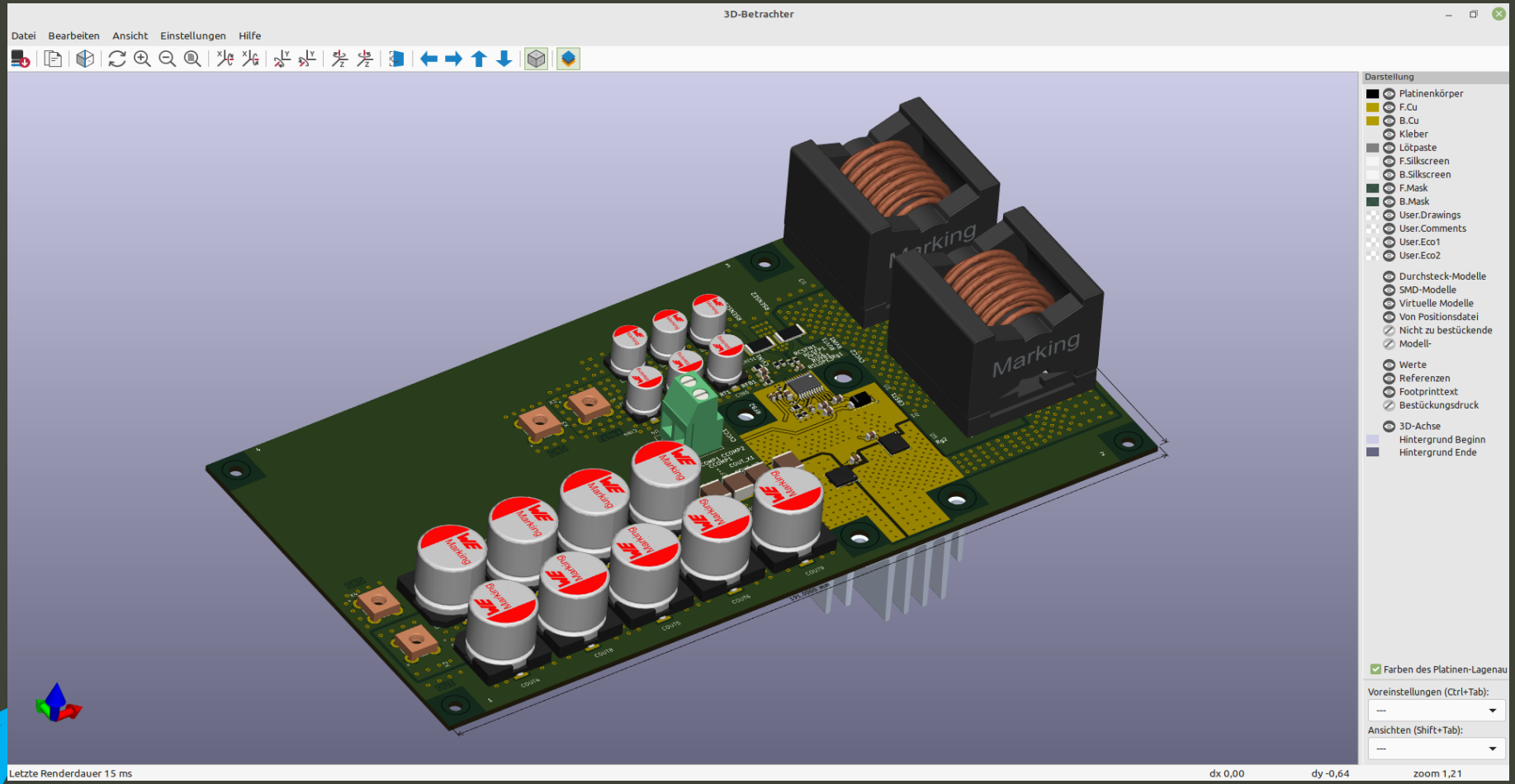
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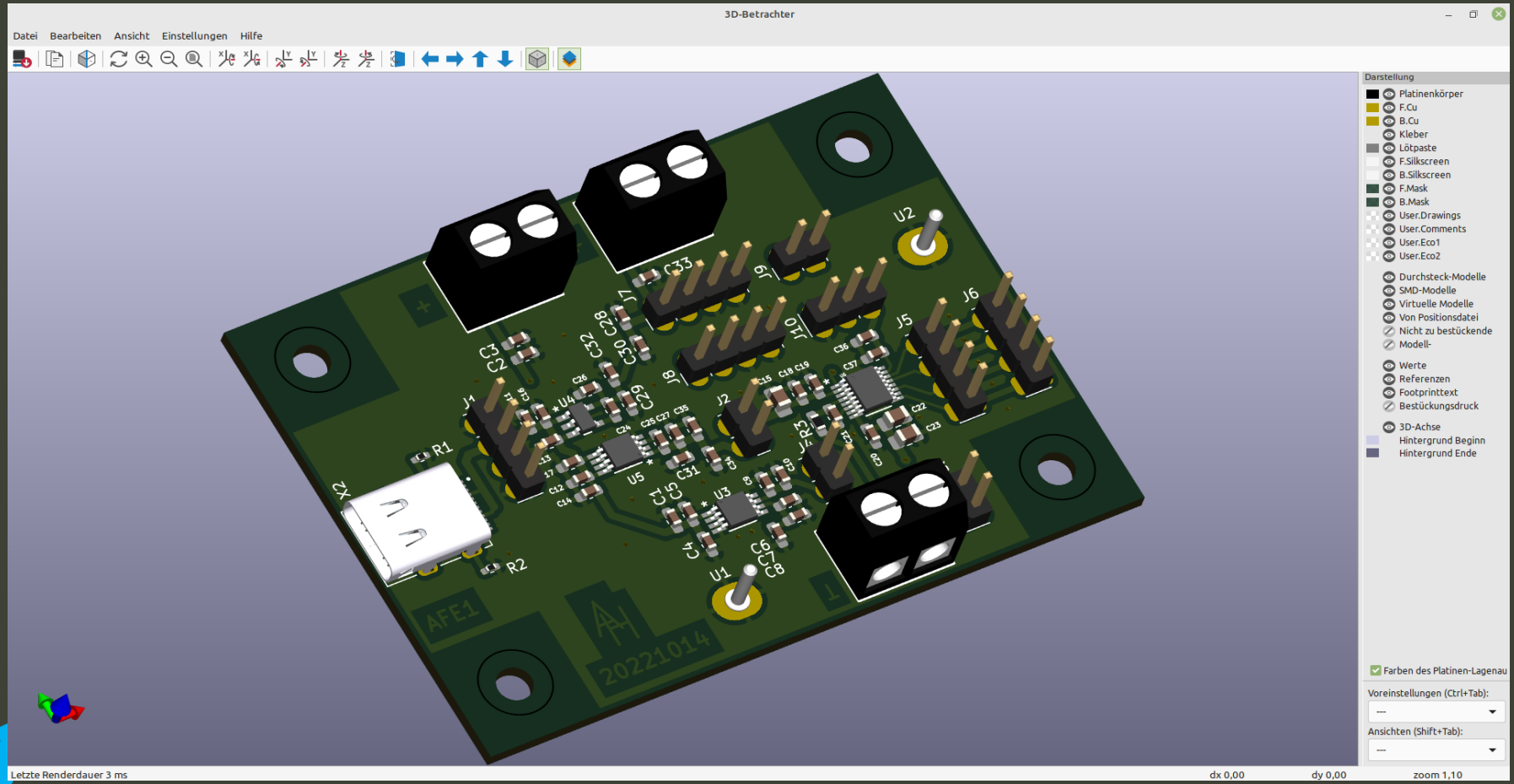
Project Seminar – What is KiCad ? (6/8)



Project Seminar – What is KiCad ? (7/8)



Project Seminar – What is KiCad ? (8/8)



Project Seminar -
What to do until next lesson ? (1/6)





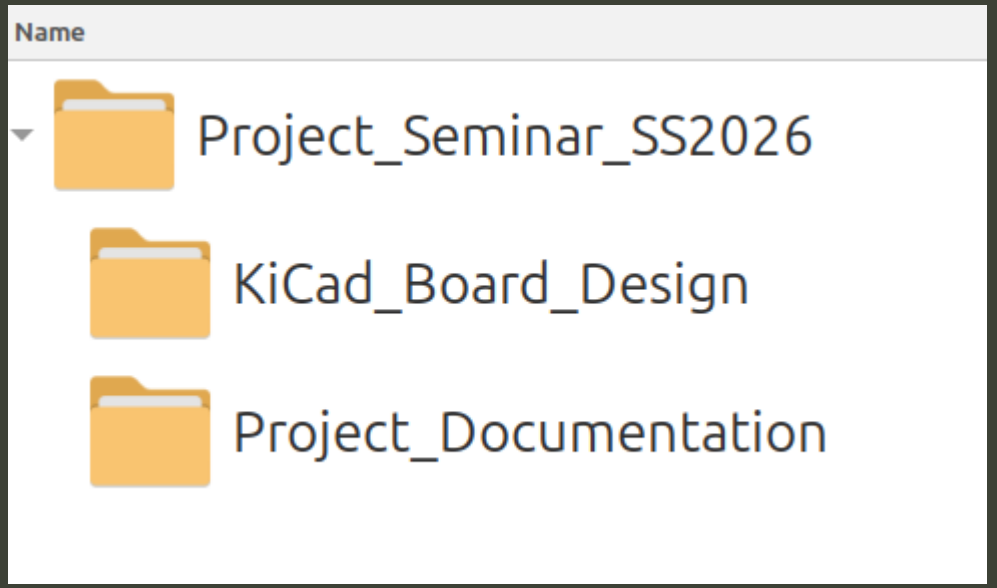
- ***KiCad EDA version 9.0.7 (or older stable version):***
  - Install KiCad on your notebook!

    https://www.kicad.org/
  - Make yourself familiar with KiCad – Read the documentation carefully ! https://docs.kicad.org/
  - Create a project main-folder Project_Seminar_SS2026 !
  - Create in the project main-folder and two sub-folders KiCad_Board_Design and Project_Documentation !



Project Seminar -
What to do until next lesson ? (2/6)

- ***Top level folder structure:***



Project Seminar -
What to do until next lesson ? (3/6)





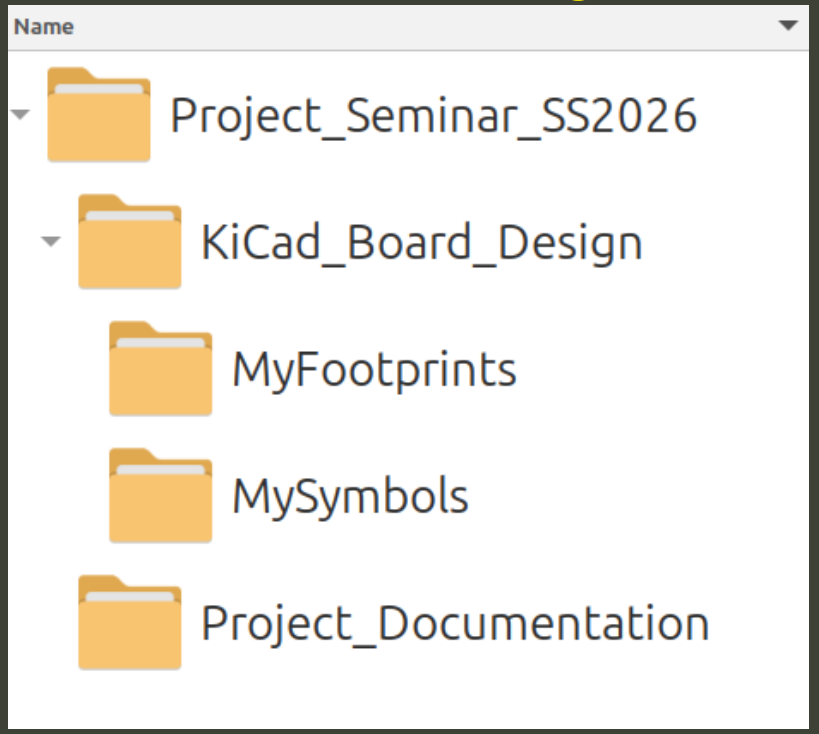
- ***Prepare a few more things:***
  - Create the following sub-folders in your KiCad_Board_Design folder with the names:
    → MySymbols
    → MyFootprints





Project Seminar -
What to do until next lesson ? (4/6)

- ***Top level folder structure → KiCad_Board_Design:***



Project Seminar -
What to do until next lesson ? (5/6)



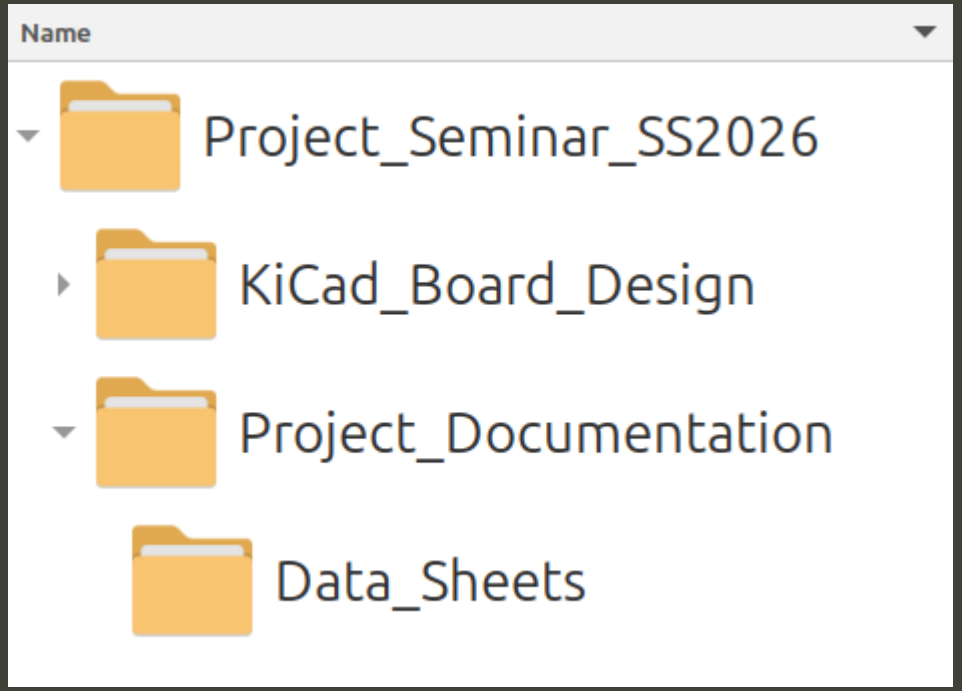


- ***Prepare a few more things:***
  - Create the following sub-folder in your Project_Documentation folder with the name:
    → Data_Sheets



Project Seminar -
What to do until next lesson ? (6/6)

- ***Top level folder structure → Project_Documentation:***



Project Seminar - SS2026



Project Seminar - Schedule (1/1)



- ***Tasks and preview:***
  - 20.03.2026: register in STUD.IP → Project Seminar
  - 20.03.2026: download and installation of KiCad 9
  - 20.03.2026: create the folder structure for project seminar
  - 20.03.2026: download KiCad 9 documentation
  - until 27.03.2026: independent study – read manuals for KiCad
  - 27.03.2026: Lesson 2 – Pcb manufacturing and pcb components

... schedule is continously updated!

Project Seminar - SS2026



Thank you !
Have a nice weekend!



